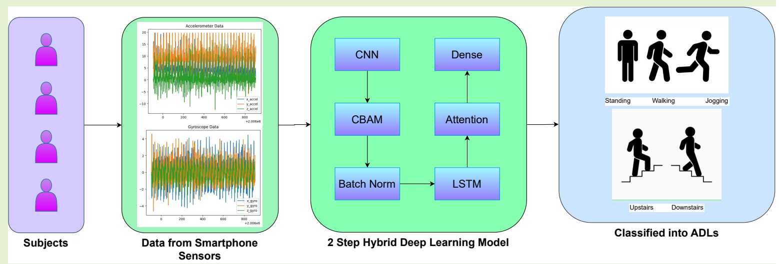
A Real-Time Deployable Attention-Driven CNN–LSTM Framework for Human Activity Recognition using Wearable Sensor

Pratibha Tokas, Vijay Bhaskar Semwal, Sr., *Member, IEEE* and Surabhi Verma

***Abstract*— Human Activity Recognition (HAR) using wearable inertial sensors is increasingly vital for healthcare, rehabilitation, and pervasive computing. Yet, many state-of-the-art models rely on handcrafted feature engineering, heavy time–frequency transformations, or large neural networks, limiting real-time deployment on resource-constrained smartphones and wearables. We propose an end-to-end, lightweight attention-guided CNN–CBAM–LSTM architecture that directly consumes raw IMU signals standardized by *StandardScaler*, eliminating costly preprocessing. Our design uniquely combines: (i) multi-resolution convolutional kernels to capture diverse motion dynamics; (ii) Convolutional Block Attention Modules (CBAM) to adaptively highlight salient spatial features and suppress sensor noise; and (iii) temporal self-attention atop stacked LSTM layers to model long-range temporal dependencies. Extensive experiments on three public HAR benchmarks—mHealth (controlled), MotionSense (uncontrolled), and UCI HAR (semi-controlled)—show that our model achieves accuracies of 99.7%, 99.5%, and 99.4% respectively, consistently surpassing recent baselines. Moreover, structured pruning and quantization compress the model to just 130.8 KB, while real-world tests on a Snapdragon 845-class Android device confirm ultra-low inference latency of 13.5 ms per segment under live streaming. This fusion of high accuracy, low latency, and minimal memory footprint demonstrates the potential of attention-driven, multi-resolution deep learning for practical, real-time edge AI in HAR applications.**



***Index Terms*— Wearable Sensor, Human Activity Recognition (HAR), Deep Learning, Inertial Measurement Unit (IMU) sensor .**

I. Introduction

HUMAN Activity Recognition (HAR) using wearable inertial sensors has gained significant traction in domains such as rehabilitation, elderly care, sports analytics, and mobile health monitoring. The advent of low-cost, portable devices like smartphones and smartwatches has enabled pervasive activity recognition in uncontrolled, real-world settings. However, achieving high classification accuracy while ensuring feasibility for deployment on resource-constrained devices remains a persistent research challenge.

Recent research has increasingly explored attention mechanisms to enhance HAR models by dynamically highlighting salient temporal patterns, sensor modalities, or feature channels [1]–[3]. These methods demonstrate strong classification performance across various public benchmarks and underline the value of attention-based architectures for sensor-based activity recognition.

Traditionally, HAR systems have depended on handcrafted feature engineering. For instance, Ghafoor et al. [4] designed a multimodal pipeline incorporating spectral features such as MFCCs, spectral kurtosis, and normalized pitch frequency alongside autocorrelation-based descriptors to improve activity classification. While effective in controlled environments, these methods demand domain expertise, involve costly signal transformations (e.g., FFT, DCT), and often introduce high latency and memory overhead, making them less suitable for real-time systems.

To overcome these limitations, deep learning approaches capable of automatic feature extraction have become popular. Some methods transform raw sensor sequences into time–frequency representations using STFT, CWT, or HHT, which are then processed by 2D or 3D CNNs for hierarchical feature learning [5], [6]. While such transformations can capture rich temporal–spectral patterns, they substantially increase computational cost and preprocessing complexity, hindering deployment on edge devices. Other approaches, such as QCLHAR [7], have explored quantum variational circuits

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to handle label scarcity; however, these remain impractical due to current quantum hardware limitations.

Signal denoising and restoration have also been explored to improve robustness. Techniques such as guided filtering, low-pass Butterworth filters [8], and Kalman filter-based smoothing [9] help reduce sensor noise and enhance signal quality. For example, [8] integrates confidence-aware Normalized Convolution and edge-preserving filtering, which contribute to improved classification accuracy. While the approach is described as suitable for real-time deployment, the use of such sophisticated preprocessing techniques may require careful optimization to ensure consistent latency performance on edge devices.

Rule-based frameworks such as Multi-EARF [10] use hierarchical clustering and fuzzy logic to improve interpretability but suffer from scalability issues and high computational overhead. Similarly, architectures like UC Fusion [11] merge sensor-specific and common features to capture cross-sensor relationships, yet introduce structural complexity and overlook long-range temporal dynamics. Tokas et al. [12] proposed an EEMD-based denoising pipeline to reconstruct sEMG signals before deep ensemble classification, improving accuracy but further increasing processing time and complexity.

Hybrid deep learning models combining convolutional and recurrent layers have also been investigated. Choudhary et al. [13], [14] integrated CNN and LSTM blocks to capture local spatial and sequential features, achieving high accuracy. However, these models lack explicit multi-resolution spatial encoding and modality-specific attention, which limits adaptability to varying motion intensities and sensor placements.

Recent work has further explored different architectures and strategies to balance performance and efficiency. Kumar et al. [15] introduced a hybrid deep reinforcement learning model (GAC) combined with cyclic GANs, achieving 98.7% and 98.4% accuracy on UCI-HAR and MotionSense datasets, respectively. Despite these impressive results, the model demands large memory, GPUs, and complex hyperparameter tuning, and it has only been tested on controlled or semi-controlled datasets. Zebhi et al. [5] proposed transforming raw sensor signals into 3D time–frequency volumes via STFT and CWT, then applying a two-stream 3D-CNN to fuse these volumes, achieving up to 99.65% accuracy; however, this comes at the cost of high preprocessing and computational demands. Lai et al. [6] fused FFT, CWT, and HHT representations using a hybrid CNN with squeeze-and-excitation and residual blocks, reporting accuracies up to 96%. Still, the method requires complex transformations and lacks explicit attention mechanisms, increasing the risk of overfitting.

Other lightweight architectures have aimed to reduce complexity by operating directly on raw sensor data. Chowdhary et al. [16] designed a Conv1D-based CNN–LSTM model that achieved 98% accuracy without preprocessing or augmentation; however, it lacks multi-resolution encoding and attention layers, potentially limiting robustness to noise. Chandramouli et al. [17] combined CNN–BiLSTM with undersampling to address class imbalance, reaching 98.5% accuracy on mHealth. Yet, the method relies mainly on data balancing rather than architectural enhancements and has not been validated for real-time deployment. Yang et al. [18] proposed MFCANN, a multi-branch convolutional network with local and global attention, achieving strong results across benchmarks, but the absence of explicit temporal modeling and high computational complexity restricts edge deployment. Similarly, Ren et al. [7] introduced a quantum contrastive learning framework that achieved 95.3% accuracy on MotionSense, but its dependence on quantum computing limits practicality. Moreover, Multi-EARF [10] by Geng et al. combined rule mining and evidential reasoning to handle uncertainty from sensor displacement, achieving 93.8% on mHealth, yet the complexity of rule mining and belief fusion constrains scalability. Finally, in [19] a stacking-based ensemble framework was proposed, achieving high accuracies (99.49% on mHealth and 96.0% on UCI-HAR), yet it requires training and maintaining multiple large base models and a meta-learner, which increases memory footprint and inference time, making it less suited for real-time use on smartphones and wearables.

Despite high reported accuracies, most existing HAR systems share common limitations: dependence on heavy preprocessing or handcrafted rules, lack of multi-resolution feature extraction, absence of explicit attention mechanisms, and high computational cost, which together impede real-time deployment on resource-constrained devices. These gaps motivate our proposed lightweight multi-resolution CNN–LSTM architecture with integrated attention, designed to enhance robustness, reduce computational overhead, and enable accurate HAR in real-world scenarios.

To bridge these gaps, our work systematically integrates four complementary design choices: (i) multi-resolution convolution kernels (3, 5, 7) to capture motion features across different scales; (ii) CBAM to adaptively recalibrate spatial features and emphasize salient sensor signals; (iii) temporal self-attention on top of stacked LSTM layers to model long-range dependencies; and (iv) structured pruning and quantization to create a lightweight, deployable model with measured inference latency and compact size. Together, these contributions enable our model to achieve consistently higher accuracy above 99% across UCI-HAR, mHealth, and MotionSense datasets—specifically 99.7% on mHealth, 99.5% on MotionSense, and 99.4% on UCI-HAR—while remaining suitable for real-time deployment on smartphones and wearables.

We rigorously evaluate our method across multiple public HAR benchmarks, including mHealth [20], MotionSense [21], and UCI HAR [22] datasets, spanning both controlled and uncontrolled environments. Without relying on any domain transformations or denoising filters, our model achieves an average accuracy of 99.5% under 5-fold stratified cross-validation, matching or surpassing complex pipelines. These results highlight the efficacy of hierarchical attention-guided learning even when operating on raw, noisy signals—thereby improving robustness and bridging the gap between accuracy and deployment feasibility.

The main contributions of this work are as follows:

- **A unified CNN–CBAM–LSTM framework for wearable sensor-based HAR:** We propose an end-to-end deep learning architecture that seamlessly combines multi-scale convolutional encoding, dual-stage attention, and sequential modeling to capture both spatial and temporal dynamics in inertial sensor data.

- **Minimal preprocessing pipeline:** The proposed model operates on statistically normalized signals using only *StandardScaler*, eliminating the need for signal denoising, handcrafted feature engineering, or domain-specific filtering, thereby enabling a streamlined pipeline that works well on raw signal.
- **Multi-resolution spatial feature enhancement via attention:** The CNN component employs parallel convolutional layers with varying kernel sizes (3, 5, and 7) along with Convolutional Block Attention Modules (CBAM) to emphasize salient features across different motion scales and sensor modalities.
- **Model compression for edge deployability:** Structured pruning and post-training quantization were applied to reduce model complexity and memory footprint. The resulting hybrid-precision model achieved a size of 130.8 KB and an average inference time of 13.5 milliseconds per segment under live streaming on a Snapdragon 845-class Android device, demonstrating its feasibility for real-time deployment in resource-constrained edge environments without degrading classification performance.
- **Comprehensive validation across diverse environments:** The model is evaluated on three public datasets—mHealth (controlled), MotionSense (uncontrolled), and UCI HAR (semi-controlled)—achieving an average accuracy of 99.5% under stratified 5-fold cross-validation, confirming its robustness and generalizability across sensing platforms, activities, and subject populations.

The remainder of this paper is structured as follows. Section II details the proposed model architecture. Section III outlines dataset characteristics, preprocessing steps, and training setup. Section IV presents performance comparisons, ablation studies, model compression, inference time discussion on domain shift and limitations. Section V concludes with future research directions.

II. Proposed Methodology

The proposed architecture integrates convolutional layers with a Convolutional Block Attention Module (CBAM) and LSTM units augmented with self-attention to effectively capture both spatial features and temporal dynamics inherent in multivariate time-series data, as illustrated in Figure 1. The model is organized into two primary modules: the CNN–CBAM Block (Figure 2) and the LSTM–Attention Block (Figure 3).

A. CNN-CBAM Block

1) Convolutional Block: The model starts with a stack of 1D convolutional layers, well-suited for processing temporal sequences such as wearable sensor data. These layers capture localized dependencies through sliding windows, with shallower layers detecting fine-grained temporal patterns and deeper layers learning abstract, high-level representations. This architecture employs convolutional filters with kernel sizes

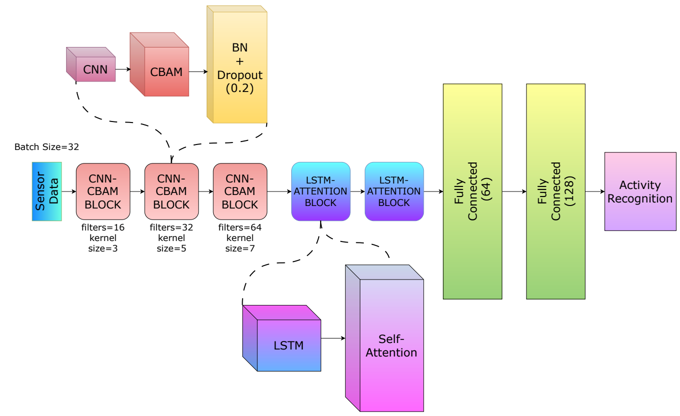

Fig. 1: Proposed CNN-CBAM-LSTM hybrid architecture

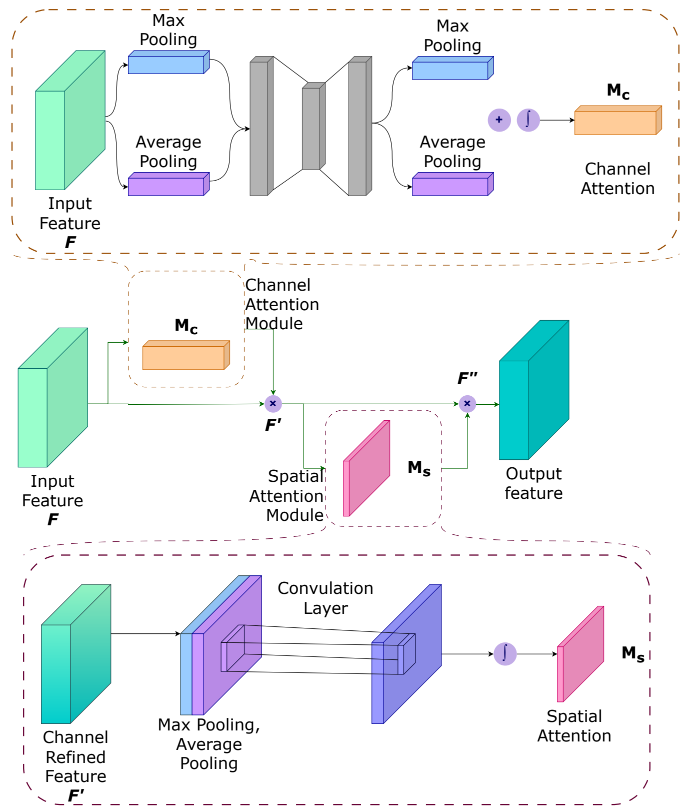

Fig. 2: Convolutional Block Attention Module (CBAM)

of 3, 5, and 7, and corresponding filter depths of 16, 32, and 64. Smaller kernels focus on short-term dynamics, while larger kernels capture broader signal trends. The convolutional operation for each layer is defined as Eq. 1:

$$y(t) = \sum_{i=0}^{K-1} x(t+i)\, w(i) + b \tag{1}$$

Where:

- $y(t)$: Output of convolution at time step t,
- $x(t+i)$: Input signal window,
- $w(i)$: Filter weight,
- b: Bias term,
- K: Kernel size.

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2) Convolutional Block Attention Module (CBAM): To enhance feature discrimination, each convolutional layer is followed by a CBAM block, which sequentially applies attention along both channel and spatial dimensions. CBAM integrates two sub-modules—channel attention and spatial attention—designed to improve the model's focus on relevant features while suppressing background noise.

Channel Attention: This module identifies the most informative channels by applying global average and max pooling across the spatial domain, feeding the resulting descriptors into a shared Multilayer Perceptron (MLP). The MLP comprises a dimensionality-reducing dense layer with ReLU activation, followed by a restoring dense layer with sigmoid activation. The resulting weights are used to rescale the original input through element-wise multiplication.

Spatial Attention: Subsequently, spatial attention highlights significant temporal regions within the channel-refined feature map. It performs average and max pooling along the channel axis, concatenates the outputs, and passes them through a convolutional layer with a 3×1 kernel and sigmoid activation to produce the spatial attention map.

The complete CBAM transformation on an intermediate feature map $F \in \mathbb{R}^{C \times H \times W}$ involves generating a 1D channel attention map $M_c \in \mathbb{R}^{C \times 1 \times 1}$ and a 2D spatial attention map $M_s \in \mathbb{R}^{1 \times H \times W}$, as shown in Eq. 2:

$$\begin{aligned} F' &= M_c(F) \otimes F, \\ F'' &= M_s(F') \otimes F', \end{aligned} \tag{2}$$

Channel Attention Module (CAM):

$$M_c(F) = \sigma\Big(\mathrm{MLP}\big(\mathrm{AvgPool}(F)\big) + \mathrm{MLP}\big(\mathrm{MaxPool}(F)\big)\Big) \tag{3}$$

Spatial Attention Module (SAM):

$$M_s(F') = \sigma\left(f^{7\times 7}\left([\mathrm{AvgPool}(F'); \mathrm{MaxPool}(F')]\right)\right) \tag{4}$$

This sequential attention mechanism enables CBAM to selectively amplify critical spatial and channel-wise features with minimal computational overhead, thereby improving the model's robustness and interpretability in noisy real-world HAR settings.

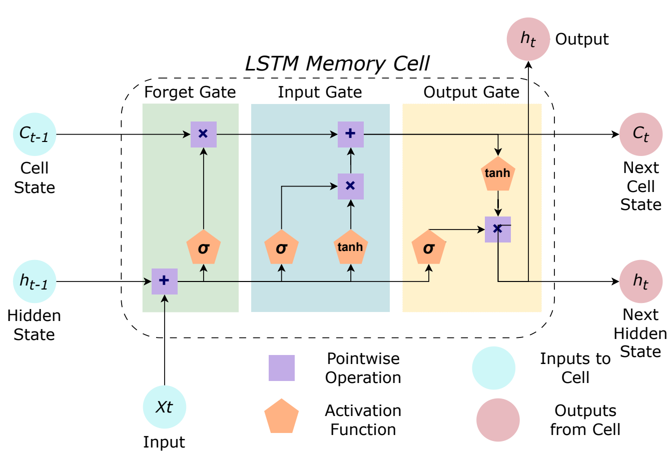

Fig. 3: Architecture of an LSTM memory cell

B. LSTM-Attention Block

1) Long Short-Term Memory (LSTM) Layers: Figure 3 illustrates the internal architecture of a Long Short-Term Memory (LSTM) cell, which addresses the limitations of traditional recurrent neural networks (RNNs) by employing gated mechanisms to regulate the flow of information. The LSTM cell comprises three primary gates: the forget gate, the input gate, and the output gate.

Forget Gate: This gate determines which parts of the previous cell state C_{t-1} should be retained or discarded. It takes the previous hidden state h_{t-1} and the current input x_t, concatenates them, and passes the result through a sigmoid activation function:

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right) \tag{5}$$

The output f_t is multiplied element-wise with C_{t-1} to filter the memory content.

Input Gate: This gate decides what new information to add to the cell state. It consists of two components: a sigmoid layer i_t that decides which values to update, and a tanh layer $\tilde{C}_t$ that creates new candidate values:

$$i_t = \sigma\left(W_i \cdot [h_{t-1}, x_t] + b_i\right) \tag{6}$$

$$\tilde{C}_t = \tanh\left(W_C \cdot [h_{t-1}, x_t] + b_C\right) \tag{7}$$

The new cell state is then updated as:

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \tag{8}$$

Output Gate: This gate determines the next hidden state h_t, which is also the output of the LSTM cell:

$$o_t = \sigma\left(W_o \cdot [h_{t-1}, x_t] + b_o\right) \tag{9}$$

$$h_t = o_t * \tanh(C_t) \tag{10}$$

This gated architecture allows the LSTM to effectively capture long-range dependencies and avoid issues such as vanishing gradients, making it particularly suitable for sequential data modeling in tasks like Human Activity Recognition (HAR).

2) Self-Attention: To enhance temporal reasoning beyond local dependencies, a self-attention mechanism is integrated after the stacked LSTM layers. While LSTMs are adept at modeling sequential patterns, their bias toward recent time steps can limit long-range dependency capture, especially in complex or cyclic activity sequences.

The self-attention module addresses this by dynamically reweighting each time step's contribution based on its contextual relevance within the entire sequence. This allows the model to focus on temporally distant yet informative segments irrespective of their position.

Given a sequence of LSTM hidden states $\mathbf{X} \in \mathbb{R}^{T \times D}$, where T is the number of time steps and D is the feature dimensionality, self-attention projects $\mathbf{X}$ into queries $\mathbf{Q}$, keys $\mathbf{K}$, and values $\mathbf{V}$ via learnable weights:

$$\begin{aligned}\mathbf{Q} &= \mathbf{X}\,\mathbf{W}_Q,\\ \mathbf{K} &= \mathbf{X}\,\mathbf{W}_K,\\ \mathbf{V} &= \mathbf{X}\,\mathbf{W}_V,\end{aligned} \tag{11}$$

The attention output is computed using scaled dot-product attention as defined by Vaswani et al. [23]:

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\,\mathbf{K}^{\mathsf{T}}}{\sqrt{d_k}}\right)\mathbf{V} \tag{12}$$

Here, d_k is the dimensionality of the key vectors, used to stabilize gradients.

Eqs. (11) and (12) show how the attention module lets each time step incorporate context from the full sequence. This combination of LSTM and self-attention enables the architecture to effectively model complex temporal dynamics with delayed transitions or long-term interdependencies. The complete pipeline is summarized in Algo. 1.

The self-attention mechanism dynamically reweights each time step's hidden state h_t by computing attention scores that reflect its contextual importance relative to all other time steps. These scores are softmax-normalized, so more relevant time steps receive higher weights, while less informative ones are suppressed. The final attention output is thus a weighted sum over the sequence, effectively concentrating the model's representational capacity on critical motion segments (e.g., sudden transitions or peaks), rather than treating all time steps equally.

This dynamic reweighting improves efficiency in two ways: (i) it enhances feature discrimination by amplifying salient temporal patterns crucial for activity classification, reducing reliance on uniformly large models; and (ii) it prevents the model from overfitting to noisy or redundant parts of the sequence by adaptively focusing computation on informative regions. In practice, the added computation from attention (a single matrix multiplication and softmax) is negligible compared to the LSTM, so overall latency remains low while accuracy benefits significantly.

Algorithm 1 Proposed Methodology

```
Input: Raw sensor data X, Activity labels Y
Output: Predicted probabilities Ŷ
procedure DATAPREPARATION
    Remove unwanted columns and NaNs
    Segment data using sliding window:
        Segment_i = X[i : i + w], i = 0, s, 2s, ...
    Label: y_i = mode(y_{i:i+w})
procedure CNNLAYER(X)
    C(x) ← σ(W ∗ x + b)
    C = {c_1, c_2, ..., c_F}
procedure CBAMATTENTION(C)
    M_c(F) ← σ(MLP(AvgPool(F)) + MLP(MaxPool(F)))
    F′ ← M_c(F) · F
    M_s(F′) ← σ(f^{7×1}([AvgPool(F′); MaxPool(F′)]))
    F″ ← M_s(F′) · F′
procedure LSTMLAYER(Fdoubleprime)
    Initialize h_0, c_0
    for t = 1 to T do
        f_t = σ(W_f[h_{t−1}, x_t] + b_f)          ▷ Forget Gate
        i_t = σ(W_i[h_{t−1}, x_t] + b_i)          ▷ Input Gate
        o_t = σ(W_o[h_{t−1}, x_t] + b_o)          ▷ Output Gate
        c̃_t = tanh(W_c[h_{t−1}, x_t] + b_c)       ▷ Cell Update
        c_t = f_t · c_{t−1} + i_t · c̃_t           ▷ New Cell State
        h_t = o_t · tanh(c_t)                     ▷ New Hidden State
procedure SELFATTENTION(H)
    Attention(Q, K, V) = softmax(QK^T / √d_k) V
procedure CLASSIFIER(Z)
    Apply: GAP → Dense(64) → Dropout → Dense(128)
    Output layer: ŷ = softmax(Wz + b)
procedure TRAINMODEL
    Loss: Sparse Categorical Crossentropy
    Optimizer: Adam (α = 0.001)
    Epochs: 50, Batch size: 32
```

III. Experiments

A. Datasets Description

To evaluate the generalization performance of the proposed model, we utilize three publicly available benchmark datasets: *mHealth*, *MotionSense*, and *UCI HAR*. These datasets were selected for their diversity in sensor modality, body placement, number of subjects, activity coverage, and recording environments—ranging from controlled lab setups to real-world scenarios.

1) **mHealth Dataset:** The mHealth (Mobile Health) dataset [20] comprises data from 10 subjects (8 males, 2 females) performing 12 physical activities, including walking, running, standing, sitting, lying, cycling, and upper-limb motions. Data were collected using wearable sensors placed on the chest, right wrist, and left ankle. Each sensor provided tri-axial accelerometer, gyroscope, and magnetometer data sampled at 50 Hz. The dataset was captured in a controlled clinical setting, making it ideal for evaluating models under noise-free conditions with multiple sensor placements.
2) **MotionSense Dataset:** MotionSense [21] includes data from 27 participants performing 6 activities of daily living: walking, jogging, ascending stairs, descending stairs, sitting, and standing. The dataset was collected using an iPhone 6s placed in the participant's front pants pocket, which recorded tri-axial accelerometer and gyroscope signals at 50 Hz. As a smartphone-based dataset captured in an uncontrolled environment, it offers realistic variability in execution patterns and sensor orientation.
3) **UCI HAR Dataset:** The UCI Human Activity Recognition (HAR) dataset [22] contains recordings from 30 healthy volunteers (aged 19–48) performing 6 common activities: walking, walking upstairs, walking downstairs, sitting, standing, and lying down. The data were acquired using a waist-mounted Samsung Galaxy S II

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smartphone, capturing tri-axial accelerometer and gyroscope signals at 50 Hz. The use of a smartphone introduces variability, making it semi-controlled and suitable for mobile HAR evaluation.

Together, these datasets span controlled (mHealth), semi-controlled (UCI HAR), and uncontrolled (MotionSense) environments, and involve different sensor types and placements. This diversity facilitates a comprehensive evaluation of the proposed architecture's ability to generalize across devices, activities, and user populations.

TABLE I: Overview of the Datasets Used

Dataset	#Sub.	Sensor	Placement	Env.
mHealth [20]	10	Acc, Gyro, Mag	Chest, Wrist, Ankle	Controlled
MotionSense [21]	27	Smartphone IMU	Front Pocket	Uncontrolled
UCI HAR [22]	30	Smartphone IMU	Waist	Semi-Controlled

B. Data Preprocessing

To prepare the raw inertial sensor data for model input, a streamlined preprocessing and segmentation pipeline was implemented. The goal was to assess the model's ability to learn meaningful representations from minimally processed data, thereby reducing dependence on complex domain-specific signal processing techniques.

Normalization: The raw multivariate time-series data were first standardized using the *StandardScaler* from the Scikit-learn library. This transformation subtracts the mean and scales each feature to unit variance, ensuring uniform feature contribution during training and promoting stable convergence. No additional operations such as denoising, outlier removal, or handcrafted feature extraction were applied. The intent behind this minimalist approach was to evaluate the model's ability to generalize from normalized, unfiltered data without relying on domain-specific signal transformations.

Segmentation: Following normalization, the continuous time-series data were segmented using a fixed-length sliding window. Each segment consisted of 100 consecutive time steps (equivalent to 2 seconds at a 50 Hz sampling rate), with a stride of 50 time steps, resulting in a 50% overlap. This overlapping segmentation increases the number of training samples while preserving temporal continuity across activity transitions. The activity label for each window was determined by the statistical mode of labels within the window. The sliding-window segmentation details have been mentioned in Table II.

TABLE II: Sliding Window Configuration for All Datasets

Dataset	Window Length (samples)	Overlap Rate (%)
mHealth [20]	100	50
UCI HAR [22]	128	50
MotionSense [21]	100	50

The hyperparameter settings used to train the proposed model are summarized in Table III.

TABLE III: Summary of architecture and training hyperparameters for the proposed model.

Component	Details
Conv Layers	Filters: [16, 32, 64], Kernels: [3, 5, 7], ReLU activation
Attention Module	CBAM (Channel + Spatial), Residual Add, Self-Attention
LSTM Layers	2 × LSTM (64 units), sequence output preserved
Fully Connected	Dense(64) → Dropout(0.2) → Dense(128) → Softmax(12)
Regularization	BatchNorm after each CNN block, Dropout = 0.2
Optimizer	Adam (learning rate = 0.001)
Loss Function	Sparse categorical crossentropy
Training Config	Epochs: 50, Batch size: 32, 10% validation split
Callbacks	EarlyStopping, ModelCheckpoint
Segmentation	Window: 100 , stride: 50
Evaluation	5-fold stratified cross-validation

IV. Results and Discussion

This section presents a detailed evaluation of the proposed CNN–CBAM–LSTM–Attention model. We examine training dynamics, compare performance against state-of-the-art methods, analyze class-wise metrics, and identify residual challenges through error analysis. All experiments are conducted on three widely used inertial datasets—mHealth [20], MotionSense [21] and UCI-HAR [22]—under consistent conditions.

A. Training Dynamics and Generalization

To assess model convergence and generalization capability, training and validation accuracy curves were monitored across epochs. As shown in Fig. 4 and Fig. 5, both accuracy curves exhibit smooth and monotonic improvement, with minimal overfitting. The tight gap between training and validation performance confirms the model's robustness and its ability to generalize well to unseen data. These results affirm the effectiveness of our design in capturing complex spatial-temporal patterns across diverse human activities.

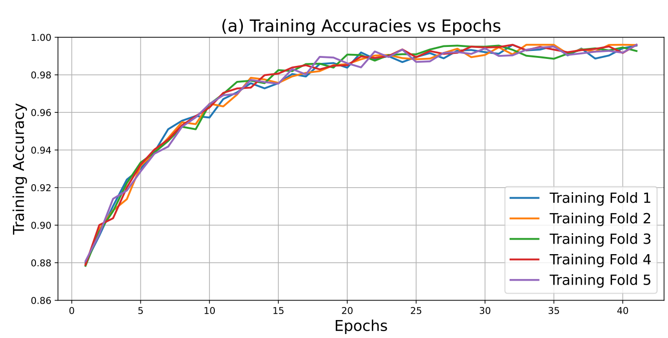

Fig. 4: Training Accuracy vs. Epochs on the MotionSense Dataset

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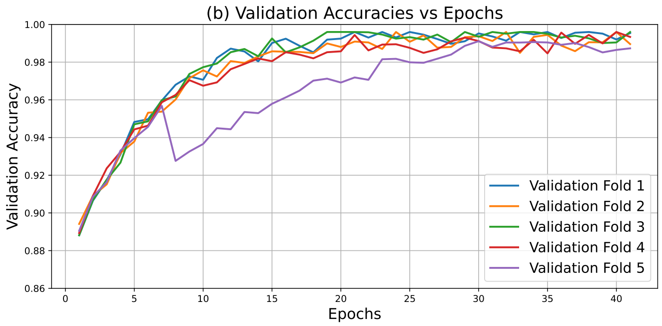

Fig. 5: Validation Accuracy vs. Epochs on the MotionSense Dataset

TABLE IV: Class-wise Performance Metrics on mHealth Dataset

Activity	Accuracy	Precision	Recall	F1-score
Standing still	99.6	97.9	97.6	97.7
Sitting and relaxing	99.6	98.1	97.7	97.9
Lying down	99.7	98.1	98.5	98.3
Walking	99.7	98.2	98.2	98.2
Climbing stairs	99.7	98.9	97.6	98.2
Waist bends forward	99.8	99.1	98.2	98.7
Frontal elevation of arms	99.8	98.5	98.8	98.6
Knees bending (crouching)	99.8	99.0	98.3	98.6
Cycling	99.7	97.8	98.9	98.3
Jogging	99.7	99.2	97.5	98.3
Running	99.7	97.1	99.4	98.2
Jump front & back	99.8	94.8	97.1	95.9

B. Class-wise Performance Analysis

To thoroughly evaluate the proposed model's effectiveness across various activity categories, we conducted a class-wise analysis using precision, recall, and F1-score on the mHealth, MotionSense, and UCI-HAR datasets. These metrics provide a more granular understanding of the model's ability to distinguish between classes and balance true positive predictions against false positives and false negatives.

To ensure unbiased evaluation and address the potential effects of class imbalance, we employed stratified 5-fold cross-validation, which maintains the class distribution in each fold and minimizes sampling bias during both training and validation. Notably, the integration of attention mechanisms—specifically the Convolutional Block Attention Module (CBAM) and self-attention layers—enables the network to adaptively emphasize salient temporal-spatial features. This attentional recalibration improves the model's capacity to identify underrepresented activity patterns, thereby enhancing classification robustness even in imbalanced scenarios.

On the *mHealth* dataset [20], the proposed model exhibited consistently strong performance across all 12 activity classes, with F1-scores for most classes exceeding 99% as shown in Table IV. Static activities such as *Standing still*, *Sitting and relaxing*, and *Lying down* were identified with near-perfect accuracy. Dynamic movements like *Cycling*, *Running*, and *Jump front/back* also achieved high classification scores, underscoring the model's ability to handle rapid transitions and complex motion patterns. A slight drop in recall was observed for *Jogging* due to overlap with similar high-intensity movements, yet the overall F1-score remained robust. These results confirm the model's ability to generalize across both postural and motion-intensive activities.

As shown in Table V, on the *MotionSense* dataset [21]—which includes fewer but diverse activity categories—the proposed model achieved uniformly high performance. Static postures such as *Sitting*, *Standing*, and *Walking* were classified with near-perfect precision and recall, demonstrating the model's robustness in recognizing low-mobility activities across different subjects. Dynamic and transitional activities like *Jogging*, *Downstairs*, and *Upstairs* presented slightly more variability in recall, particularly *Jogging* (F1-score: 98.7), likely due to overlapping movement patterns with similar high-intensity activities. Nonetheless, all classes recorded F1-scores above 98.0, indicating consistent performance across varying movement intensities and directions. These results underscore the model's strong generalization capability across both postural and locomotor activities.

TABLE V: Per-Class Evaluation Metrics on the *MotionSense* Dataset

Activity	Accuracy	Precision	Recall	F1-score
Sitting	99.4	99.2	98.9	99.0
Standing	99.5	99.2	98.8	99.0
Walking	99.5	98.7	98.7	98.7
Jogging	99.4	96.3	96.3	96.3
Upstairs	99.4	96.1	96.7	96.4
Downstairs	99.7	96.5	99.1	97.8

On the *UCI HAR* dataset [22]—which features a balanced mix of postural and transitional activities—the proposed model maintained consistently high classification performance across all activity classes as shown in Table VI. Static postures such as *Sitting*, *Standing*, and *Laying* were detected with remarkable precision and recall, each recording F1-scores above 99.3, indicating minimal confusion between closely related stationary activities. Locomotor actions such as *Walking*, *Upstairs*, and *Downstairs* also yielded robust results, with F1-scores ranging from 99.1 to 99.4. Notably, the model performed uniformly well across activities involving similar kinematic patterns—e.g., *Walking Upstairs* vs. *Walking Downstairs*—demonstrating its fine-grained temporal sensitivity. Overall, these results reinforce the model's strong capacity to generalize across both repetitive and static activity types within a semi-controlled smartphone-based sensing environment.

TABLE VI: Per-Class Evaluation Metrics on the *UCI HAR* Dataset

Activity	Accuracy	Precision	Recall	F1-score
Walking	99.6	98.7	99.0	98.9
Walk_Upstairs	99.1	98.5	96.1	97.3
Walk_Downstairs	99.3	98.1	98.3	98.2
Sitting	99.4	98.0	98.4	98.2
Standing	99.3	97.8	97.1	97.5
Laying	99.5	97.5	99.6	98.5

Overall, this class-wise evaluation highlights the proposed model's effectiveness in reliably distinguishing a diverse range

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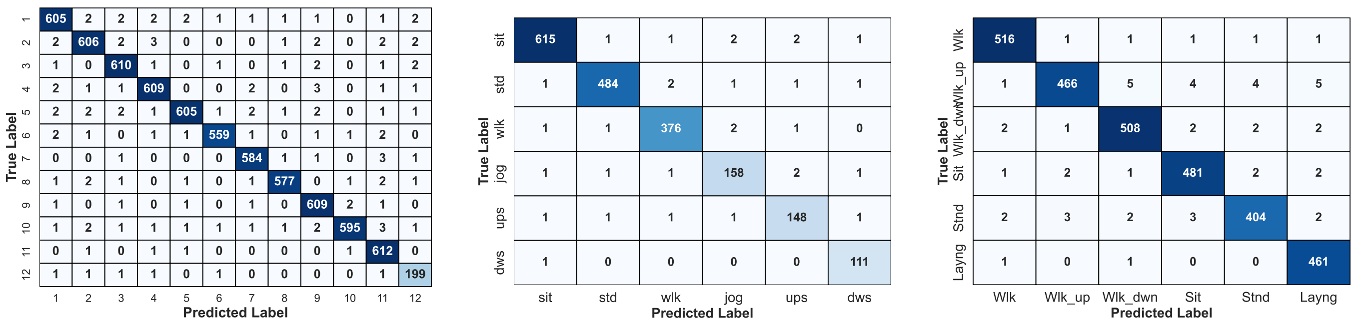

(a) Confusion Matrix for mHealth (b) Confusion Matrix for MotionSense (c) Confusion Matrix for UCI-HAR

Fig. 6: Confusion matrices for all three datasets

of human activities. The consistently high precision, recall, and F1-scores reflect its strong discriminative capacity and applicability in real-time, sensor-based HAR systems.

C. Comparison with Existing Methods

TABLE VII: Comparison with Benchmark Models on mHealth and MotionSense

Dataset	Benchmark Work	Year	Accuracy (%)
mHealth	Geng *et al.* [10]	2024	93.8
	Lai *et al.* [6]	2024	96.0
	Choudhary *et al.* [16]	2025	99.1
	Chandramouli *et al.* [17]	2024	98.5
	Varshney *et al.* [24]	2022	98.6
	Essa *et al.* [25] (CSNet)	2023	97.66
	Essa *et al.* [25] (TCCSNet)	2023	98.60
	Proposed Model	2025	**99.7**
MotionSense	Zebhi *et al.* [5]	2023	97.7
	Ren *et al.* [7]	2024	95.3
	Xia *et al.* [26]	2023	94.4
	Kumar *et al.* [15]	2025	98.8
	Luwe *et al.* [27]	2022	94.17
	Saha *et al.* [28]	2024	95.35
	Proposed Model	2025	**99.5**
UCI-HAR	Zhixuan *et al.* [18]	2024	93.2
	Yao *et al.* [29]	2024	97.0
	Gaud *et al.* [8]	2024	95.0
	Kang *et al.* [11]	2023	96.3
	Xia *et al.* [26]	2023	94.1
	Kumar *et al.* [15]	2025	98.7
	Proposed Model	2025	**99.4**

To assess the effectiveness of the proposed CNN–CBAM–LSTM–Attention architecture, a comprehensive comparison was carried out against several recent state-of-the-art (SoTA) models using the mHealth [20], MotionSense [21], and UCI-HAR [22] datasets. All models were evaluated under consistent conditions, including identical preprocessing using StandardScaler normalization, fixed-length sliding window segmentation, and 5-fold stratified cross-validation to ensure methodological uniformity.

As presented in Table VII, the proposed model achieves superior performance across all three datasets. On the mHealth dataset, it attained the highest reported accuracy of **99.7%**, outperforming prior strong baselines such as Choudhary *et al.* [16] (99.1%) and TCCSNet [25] (98.60%). Similarly, for the MotionSense dataset, the proposed approach achieved **99.5%** accuracy, exceeding the performance of Kumar *et al.* [15] (98.8%) and Zebhi *et al.* [5] (97.7%). On the UCI-HAR dataset, it recorded an accuracy of **99.4%**, surpassing Kumar *et al.* [15] (98.7%) and Kang *et al.* [11] (96.3%). These results underscore the robustness and generalizability of the proposed model across diverse sensing conditions and activity types. The corresponding performance comparison of proposed model on mHealth and MotionSense datasets are further visualized in Figures 7 and 8. Moreover, the confusion matrices for all the three datasets have been shown in Fig. 6.

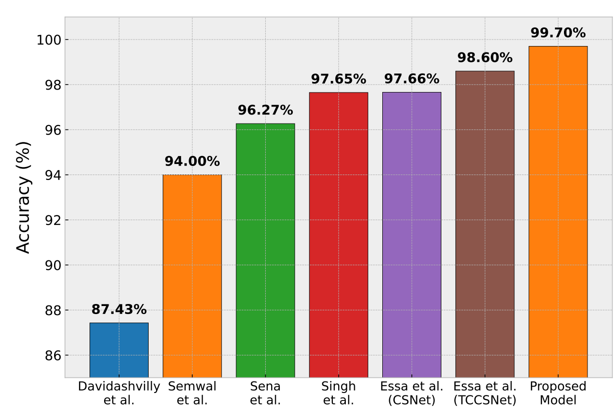

Fig. 7: Comparison of proposed work with SoTA on mHealth dataset

The consistently higher accuracies and F1-scores across all datasets can be attributed to the model's explicit integration of multi-resolution convolution, CBAM-based spatial attention, and dual-stage temporal modeling. Specifically, the multi-resolution CNN layers (with kernel sizes of 3, 5, and 7) extract motion patterns at different temporal scales, which helps capture both fine-grained and broader activity dynamics. The CBAM modules further enhance spatial representation by adaptively recalibrating channel and spatial features, enabling the model to emphasize the most informative sensor axes and local motion patterns while suppressing noise from inconsis-

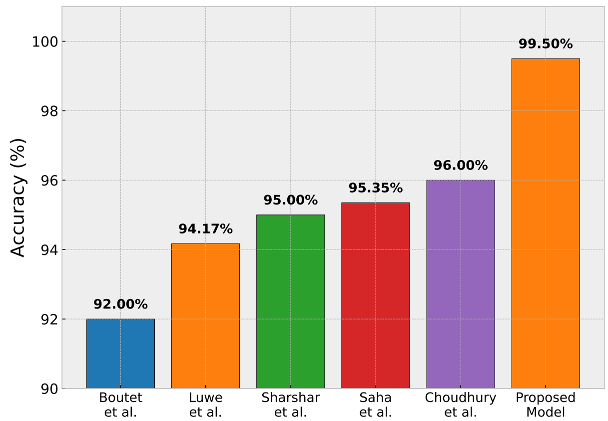

Fig. 8: Comparison of proposed work with SoTA on motion-sense dataset

tent sensor placement or individual differences.

In the temporal domain, stacked LSTM layers learn sequential dependencies, while the self-attention mechanism refines these temporal features by selectively focusing on frames that are more indicative of transitions or distinctive motion peaks. This combination helps the model remain sensitive to brief or subtle activity changes that might otherwise be overlooked.

Together, these components enable the model to dynamically prioritize the most relevant spatial and temporal features, rather than processing all sensor channels and time steps uniformly. This design contributes to greater robustness against noise, improved recognition of short or overlapping activities, and stronger generalization across datasets collected in different conditions. By comparison, baseline models that lack explicit attention mechanisms or multi-resolution convolution are less capable of capturing these finer spatial-temporal patterns, which can lead to reduced performance. The effectiveness of each component has been further elaborated in subsection IV-D.

While the proposed framework relies solely on statistical normalization using `StandardScaler` and avoids extensive preprocessing or data augmentation, its robust performance across three diverse datasets—collected in controlled, semi-controlled, and uncontrolled environments—demonstrates the model's inherent generalizability. This outcome is largely attributed to the model's architectural design, which integrates channel and spatial attention (CBAM) with temporal modeling (LSTM and self-attention) to dynamically emphasize informative features and suppress noise.

D. Ablation Study

To rigorously assess the contribution of each component in the proposed hybrid architecture, we conduct a comprehensive ablation study. This analysis is essential for understanding how spatial encoding, temporal modeling, and attention mechanisms collectively contribute to overall performance in Human Activity Recognition (HAR). The study involves systematically isolating key components—namely the CBAM, LSTM layers, and self-attention—and comparing each variant against the full model.

- **CNN–CBAM**: This variant consists of convolutional layers followed by CBAM, which recalibrates spatial features via channel and spatial attention. It excludes both LSTM layers and self-attention, thereby focusing solely on local spatial encoding.
- **LSTM–Attention**: This configuration uses stacked LSTM layers followed by self-attention for temporal modeling. It operates directly on the raw input sequence, bypassing CNN-based spatial feature extraction.
- **CNN–LSTM**: This model combines convolutional layers with LSTM units for spatiotemporal modeling, but omits both CBAM and self-attention. It serves as a conventional baseline often used in wearable HAR systems.
- **Proposed CNN–CBAM–LSTM–Attention**: This is the full model, integrating all three components—CBAM, LSTM, and self-attention—into a unified end-to-end framework.

The classification performance of each configuration is reported in Table VIII. The proposed model significantly outperforms all ablated versions, achieving an accuracy of 99.66%. The CNN–LSTM model, which lacks attention mechanisms, achieves a lower accuracy of 91.56%, confirming that attention enhances both spatial and temporal representations. Removing the LSTM layers and relying solely on CNN–CBAM yields 90.72%, suggesting that spatial attention alone is insufficient for modeling temporal dependencies. The poorest performance is observed in the LSTM–Attention configuration (82.97%), indicating that temporal modeling without early spatial feature extraction leads to degraded accuracy.

TABLE VIII: Performance Comparison of Model Variants on the mHealth Dataset

Model Variant	Accuracy (%)
CNN–CBAM	90.72
LSTM–Attention	82.97
CNN–LSTM	91.56
CNN–CBAM–LSTM–Attention	**99.66**

These results clearly demonstrate the synergistic effect of combining convolutional encoding, attention-guided spatial recalibration, temporal modeling via LSTM, and global reweighting through self-attention. While each component contributes individually, the integration of all modules leads to a marked improvement in performance. Notably, the inclusion of CBAM and self-attention enhances the model's ability to focus on salient activity features and long-range dependencies, which are critical for accurately recognizing transitional or cyclic human movements.

The ablation findings thus validate the architectural design of the proposed model and underscore the necessity of holistic spatial-temporal attention modeling for robust, real-world HAR performance.

E. Model Compression and Inference Efficiency

To evaluate the deployability of the proposed CNN–CBAM–LSTM model, structured pruning and post-training quantization were applied using the TensorFlow

Model Optimization Toolkit and TensorFlow Lite. Convolutional and dense layers were pruned and quantized to INT8, whereas the LSTM and attention modules were retained in float precision due to conversion constraints involving dynamic tensor operations. To measure inference latency on an Android device, we deployed the quantized TensorFlow Lite model into a custom Android application developed in Java. Live streaming tests were conducted on a Snapdragon 845-class device, where real-time IMU data was collected at 50 Hz and buffered into overlapping windows with 50% overlap. For each buffered segment, timestamps were recorded immediately before and after calling `interpreter.run()` using `System.nanoTime()`. The reported latency corresponds to the average over 1000 consecutive segments, excluding sensor polling and preprocessing, but including JNI and interpreter overhead. Under these real-world conditions, the compressed model achieved an average inference latency of approximately 13.5 milliseconds per segment, confirming its suitability for real-time Human Activity Recognition (HAR) on edge platforms. This confirms its suitability for real-time Human Activity Recognition (HAR) on edge platforms. Note that the reported latency reflects inference performance only and does not include potential overhead from adaptive domain calibration or personalization modules. Despite its mixed-precision configuration, the model preserved predictive performance while significantly reducing memory footprint and inference latency as shown in Table IX.

TABLE IX: Comparison of Model Variants in Terms of Size and Inference Time

Model Variant	Size (KB)	Inference Time (ms)
Baseline	∼942.00	41.8
Pruned Only	471.10	37.5
Quantized Only	∼316	16.2
Pruned + Quantized (live)	130.8	**13.5**

F. Domain Shift and Real-World Deployment Considerations

While the proposed CNN–CBAM–LSTM model shows strong accuracy across controlled (mHealth), semi-controlled (UCI HAR), and uncontrolled (MotionSense) datasets, real-world deployments inevitably face domain shift. Domain shift can arise from changes in device hardware, sensor orientation (e.g., wrist vs. waist), user gait, demographics, or unseen activity contexts, potentially leading to reduced accuracy when models trained offline are deployed on new users or environments.

To partially address this, the model incorporates CBAM and temporal self-attention, which dynamically reweight spatial and temporal features, improving resilience to moderate variability and sensor noise. However, explicit domain adaptation, transfer learning, or online personalization was not implemented in this study and is planned as future work.

G. Limitations

The current model operates solely on inertial data, which may limit performance in recognizing subtle or complex activities. Furthermore, while the attention modules enhance robustness, the model does not yet include adaptive domain recalibration, which could help mitigate performance drops under significant domain shift. These limitations highlight opportunities for multi-modal integration and real-time personalization in future work.

V. Conclusion and Future Work

This study proposed a compact and effective hybrid deep learning framework for wearable sensor-based Human Activity Recognition (HAR), integrating multi-scale Convolutional Neural Networks (CNNs), Convolutional Block Attention Modules (CBAM), stacked Long Short-Term Memory (LSTM) layers, and a dual-stage temporal attention mechanism. Designed to operate on minimally processed inertial signals standardized using only *StandardScaler*, the model eliminates the need for handcrafted features or domain-specific denoising techniques, ensuring a streamlined and deployable HAR pipeline.

Unlike traditional HAR approaches that rely on extensive preprocessing or complex feature engineering, the proposed architecture achieves strong generalization performance—reporting an average accuracy of 99.5% across three publicly available datasets: mHealth (controlled), MotionSense (uncontrolled), and UCI HAR (semi-controlled). The inclusion of CBAM-based spatial recalibration and temporal attention allows the model to effectively capture salient features and long-range dependencies, even under sensor variability, noise, or activity overlap.

To support practical edge deployment, we applied structured pruning and post-training quantization, compressing the model to a 130.8 KB format while achieving a low inference latency of 13.5 ms per segment on a Snapdragon 845-class Android device. This confirms the model's suitability for real-time, resource-constrained applications in healthcare, fitness, and ambient-assisted living.

Future work will explore unsupervised domain adaptation, lightweight continual learning, and user-specific calibration to enhance robustness under domain shift, and extend the framework to multi-modal sensors (e.g., sEMG, gyroscopes, microphones) and transformer-based temporal encoding to improve interpretability and scalability.

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